

NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY, KHULNA

Department of Computer Science & Engineering

QUALITY ASSURANCE & DEFECT TRACKING

AI Smart Campus System

Comprehensive Bug Report, Root-Cause Analysis & Verification Log

Course Code & Title	CSE4204: Software Engineering & AI Integration
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Section & Team	Section 8A · Team 07 (T07)
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Team Members	Md. Nazmus Shakib (11220320852) [Lead] Samira Akter Mitu (11220320858) Tanvin Sadik Dhrubo (11220320860) Khan Waziur Rahman (11220320861)
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Total Bugs Identified	5
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Bugs Fixed & Verified	5 (100%)
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Remaining Issues	0
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Date	September 2026
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1. Bug Summary Dashboard

5

TOTAL BUGS

1

HIGH SEVERITY

3

MEDIUM SEVERITY

5

FIXED & VERIFIED

2. Detailed Bug Tracking Records

BUG-001: Whitespace trimming and email case-normalization missing on form submission in LoginPage and RegisterPage.

MEDIUM
Verified

Module: Authentication & Validation

STEPS TO REPRODUCE:

1. Navigate to /login.
2. Enter email with trailing space or mixed case (e.g., " Student1@NUBTKHULNA.ac.bd ").
3. Click Sign in securely.

Expected Behavior: Form input should automatically trim leading/trailing whitespace and normalize to lowercase before API submission.

Actual Behavior: Whitespace resulted in raw string submission and validation error from strict server regex before normalization fix.

Resolution: Updated validateLoginForm and handleSubmit in LoginPage.jsx and RegisterPage.jsx to normalize email strings before client validation and API dispatch.

Retest Result: Verified: Whitespace and mixed case emails are now normalized seamlessly and login completes successfully (HTTP 200).

BUG-002: Printable HTML transcript rendered "N/A" with raw template artifact when student model section attribute was null.

LOW
Verified

Module: Academic Management / Export

STEPS TO REPRODUCE:

1. Authenticate as student with unassigned section.
2. Access GET /api/student/transcript.
3. Observe HTML output header.

Expected Behavior: Safe fallback to section placeholder without throwing template rendering warnings.

Actual Behavior: Section rendered without null coalescing fallback.

Resolution: Added null-safe operator and fallback \$student→section ?? "N/A" in AcademicExportController.php.

Retest Result: Verified: HTML transcript generates cleanly with proper student header information.

BUG-003: Playwright E2E configuration defaulted to downloading bundled Chromium which failed in restricted offline/Windows environments without system channel declaration.

MEDIUM

Verified

Module: Automated Testing Infrastructure

STEPS TO REPRODUCE:

1. Run `npx playwright test` in vanilla Windows terminal without downloaded playwright brows
2. Browser launch fails with Executable doesn't exist.

Expected Behavior: Test configuration should leverage installed Google Chrome (channel: chrome) directly.

Actual Behavior: Failed looking for non-existent AppData headless shell path.

Resolution: Configured channel: "chrome" and baseURL in playwright.config.js.

Retest Result: Verified: All browser and E2E tests execute cleanly using system Google Chrome (100% pass).

BUG-004: Faculty accessing academic workspace required explicit course-to-faculty ownership check to prevent unauthorized grading of other professors' courses.

HIGH

Verified

Module: Authorization & RBAC

STEPS TO REPRODUCE:

1. Log in as Faculty A.
2. Send GET `/api/academic-management/courses/{courseId}/workspace` for Course B (assigned t

Expected Behavior: API must return HTTP 403 Forbidden preventing access to unassigned course grading rosters.

Actual Behavior: Potential parameter tampering risk if course ID was manipulated.

Resolution: Implemented strict `ensureCourseAccess()` policy method in AcademicManagementController.php verifying faculty user ID matches course assignment.

Retest Result: Verified: Direct URL access to unassigned course workspace returns HTTP 403 Forbidden.

BUG-005: Student attendance CSV stream export lacked explicit quote encapsulation for course titles containing commas.

MEDIUM

Verified

Module: Academic Export / CSV Stream

STEPS TO REPRODUCE:

1. Enroll student in course with comma in title (e.g. "Software Development II: Web Development")
2. Download CSV attendance report.

Expected Behavior: CSV fields must be encapsulated with fputcsv standard quoting.

Actual Behavior: Raw unquoted strings caused column misalignment in Excel/spreadsheet viewers.

Resolution: Standardized fputcsv stream writer in AcademicExportController.php with proper RFC 4180 field enclosure.

Retest Result: Verified: CSV downloads import seamlessly with proper column structure.